PROJECT DUCA
DUCABRAIN · UBUNTU 24.04
OPENCLAW 2026.4.2 · OLLAMA
JOHNSTOWN, PA

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Autonomous Agent Architecture & AI Systems Design

01 Project Overview

Project Duca is a fully autonomous, locally-hosted AI job search pipeline running on a purpose-built Ubuntu 24.04 workstation. It scrapes job boards, ingests forwarded email alerts, verifies and enriches leads, scores them against the operator's profile via local LLM, builds per-lead application packets, and delivers a structured Morning Report to Discord. All outputs are filed to Google Drive. Zero cloud dependency. Zero paid API usage.

The system was built around two principles: absolute data sovereignty and human-in-the-loop governance. Cloud-based job tools and auto-applier bots expose sensitive career data to third-party servers while triggering ATS blacklists that silently kill applications. Duca keeps all processing on the operator's machine. The pipeline finds the signal. The operator makes every decision. Nothing ships without human review.

DEPLOYMENT

100% Local

All inference on DucaBrain. No cloud. No third-party LLM services.

COMMUNICATION

Discord Bot

Private guild. All reporting flows through a dedicated channel.

SCHEDULING

Nightly · Midnight ET

OpenClaw cron. Self-managing. Persistent across reboots.

02 Pipeline Architecture

```
STAGE 1    Scout (python-jobspy) — Web scrape from Indeed + LinkedIn
STAGE 1.3 Email ingest (Gmail) — Parse forwarded alerts (6 sender-specific parsers)
STAGE 1.4 Merge + cross-dedup — Combine sources, remove duplicates
STAGE 1.5 Vault dedup (7-day window) — Suppress previously seen leads
STAGE 2    Gatekeeper verify — URL liveness, ATS tracing, salary lookup, ghost detection
STAGE 3    Analyst (Ollama LLM) — Score leads against operator profile
STAGE 4    Drive upload — Per-lead application packets + manual-check roster
STAGE 5    Vault write — SQLite logging for all scored leads
```

Only Stage 3 uses the LLM. Everything else is deterministic Python. Orchestrated by a 530-line bash script. Duca (the OpenClaw agent) fires the pipeline via cron, parses the output, formats the Morning Report, and delivers it to Discord.

03 Technology Stack

AGENT FRAMEWORK OpenClaw 2026.4.2	LOCAL INFERENCE Ollama	SCORING MODEL qwen3.5:9b
WEB SCRAPING python-jobspy	ENRICHMENT SearXNG (localhost)	COMMUNICATION Discord Bot API
OUTPUT STORAGE Google Drive (gog CLI)	DEDUPLICATION SQLite Vault	MONTHLY COST $0

04 Critical Debugging

SILENT SYSTEM FAILURE — ROOT CAUSE ANALYSIS

Problem: OpenClaw's `api: "ollama"` mode sends system prompts using `role: "developer"`. Ollama silently drops unrecognized roles. Duca's 24,000-character system prompt vanished on every request with no error output.

Fix: Switched to `api: "openai-completions"` with an explicit compatibility block routing through `role: "system"`. Full system prompt reception confirmed on first attempt after fix.

COMMUNITY-CONTRIBUTED FIXES — OPENCLAW ISSUES #46298 & #49572

Problem: `sessions_spawn` and `sessions_yield` fail silently in OpenClaw 2026.3.13. No posted fix existed. Cross-agent messaging required two undocumented configuration gates with no mention in official documentation.

Fix: Built a working replacement using `sessions_send` with a blocking wait pattern. Published the first documented solutions to both open issues. Documented the undiscovered configuration requirements.

PRODUCTION HARDENING — 24 BUGS FIXED IN 4 NIGHTS

LLM scoring runtime reduced from 72 minutes to 8–15 minutes via direct API calls, adaptive batching, and split-retry with recursive halving. Context overflow (97KB merged JSON exceeding 32K budget) resolved by restructuring pipeline stdout. Cross-company contamination in SearXNG enrichment fixed with strong-token relevance filtering.

05 Key Accomplishments

SYSTEM DESIGN & ARCHITECTURE

- Purpose-built AI workstation (DucaBrain) configured exclusively for autonomous agentic workloads
- Multi-stage pipeline with deterministic Python stages and a single LLM scoring stage for zero-cost operation
- 6 sender-specific email parsers: LinkedIn, Indeed, Ladders (3 formats), Lensa, Virtual Vocations, MyPerfectResume

- Structured Google Drive output with per-lead packet generation and automatic operator sharing

CONFIGURATION & INTEGRATION

- OAuth 2.0 Google Workspace authentication on a headless Linux server using manual callback flow
- Discord bot integration with guild routing, channel allowlisting, and message acknowledgment
- OpenClaw multi-agent configuration with model assignments, tool deny lists, and isolated workspaces

OPEN-SOURCE CONTRIBUTIONS

- Posted first documented solutions to two open OpenClaw GitHub issues with no known public fix
- Published full technical writeup with root cause analysis and working replacement patterns

06 Technical Competencies Demonstrated

AI / LLM SYSTEMS

- Local LLM deployment and management (Ollama)
- Multi-stage pipeline architecture
- System prompt design and context engineering
- Model selection and parameter tuning
- API adapter compatibility analysis

PROBLEM SOLVING

- Silent failure diagnosis without log output
- Source code analysis for root cause identification
- Systematic backup-verify-restore protocols
- Research-first, protocol-driven methodology

INFRASTRUCTURE & INTEGRATION

- Linux server administration (Ubuntu 24.04)
- OAuth 2.0 headless authentication
- Google Workspace API integration
- Discord bot deployment
- Environment variable security

TOOLS & FRAMEWORKS

- OpenClaw 2026 — agent gateway
- Ollama — local model serving
- gog CLI — Google Workspace automation
- python-jobspy — job board scraping
- SearXNG — search and enrichment

07 Operational Status

Operational since March 2026. Rebuilt and hardened April 2026. 675+ leads processed. 24 bugs fixed. Morning Report delivered nightly.

- Gateway Running
- Discord Connected
- Drive Authenticated
- Cron Armed
- Pipeline Operational